# AWS Solutions Architect Associate (SAA-C03) Study Notes

## Topic: EC2 Advanced Networking & Features

### 1. Private vs. Public vs. Elastic IP

Understanding IP addressing behavior is critical for the exam, especially regarding instance lifecycle events (Stop/Start).

#### Private IP

- **Definition:** The internal IP address used for communication within the VPC.
- **Persistence:** Retained for the life of the instance. If you Stop and Start the instance, the Private IP **does not change**.
- **Scope:** Only reachable within the VPC or via VPN/Direct Connect (unless NAT is used).

#### Public IP

- **Definition:** An IP address reachable from the internet.
- **Behavior:**
  - Assigned automatically if `Auto-assign Public IP` is enabled in the subnet settings or requested at launch.
  - **Dynamic:** If you Stop and then Start the instance, the Public IP **will change**.
  - **Hostname:** Cannot be manually moved between instances.

#### Elastic IP (EIP)

- **Definition:** A static, fixed Public IP address allocated to your AWS account.
- **Use Case:** Masking instance failures by rapidly remaping the address to another instance.
- **Behavior:**
  - You own it until you release it.
  - **Static:** It does not change if you stop/start the instance.
  - **Scope:** Region-specific (cannot move an EIP from us-east-1 to us-west-2).
- **Cost:**
  - Free if attached to a running instance and it is the *only* EIP attached.
  - **Charges apply if:** The instance is stopped, the EIP is unattached, or you have more than one EIP attached to a running instance.

### 2. EC2 Placement Groups

Placement groups determine how instances are placed on underlying hardware to optimize for performance, durability, or availability.

| Type | Description | Use Case | Constraint |
|---|---|---|---|
| **Cluster** | Instances are packed close together inside a single Availability Zone (AZ). Provides low latency and high network throughput (10Gbps+). | High Performance Computing (HPC), tightly coupled applications, supercomputing. | **Single AZ only.** High risk of concurrent failure if the rack fails. |
| **Spread** | Instances are placed on distinct underlying hardware (different racks). Maximizes availability. | Critical applications where individual instances must be isolated from each other's hardware failure. | Max **7 instances per AZ** per placement group. |
| **Partition** | Instances are spread across logical partitions (groups of racks). Instances in one partition do not share hardware with instances in other partitions. | Distributed & Big Data workloads: Hadoop, Cassandra, Kafka. | Can span multiple AZs. Scales to 100s of instances. |

## 3. Elastic Network Interfaces (ENI) - Overview

An ENI is a logical networking component in a VPC that represents a virtual network card.

- **Scope:** Bound to a specific **Availability Zone (AZ)**. You cannot attach an ENI created in AZ A to an instance in AZ B.
- **Attributes:** An ENI can have:
  - A primary private IPv4 address.
  - One or more secondary private IPv4 addresses.
  - One Elastic IP address (per private IPv4).
  - One Public IPv4 address.
  - One or more security groups.
  - A MAC address.

### Types of ENI Attachment

1. **Primary ENI (eth0):** Created automatically when an instance is launched. **Cannot be detached** from the instance.
2. **Secondary ENI (eth1+):** Can be created independently and attached/detached from instances on the fly.

## 4. ENI - Extra Reading (Exam Tips)

- **Failover Scenarios:** You can create a low-cost failover solution by creating a secondary ENI with a private IP and an Elastic IP attached. If the primary instance fails, you can move the ENI to a standby instance.
- **Dual Homing:** You might attach two ENIs to an instance:
  - **ENI 1:** Connected to a management subnet (for SSH/RDP access).
  - **ENI 2:** Connected to a data subnet (for application traffic).
- **Security Groups:** Security Groups are attached to the **ENI**, not the instance itself. This means different ENIs on the same instance can have different security group rules.

## 5. EC2 Hibernate

EC2 Hibernate allows you to pause an instance and resume it later, saving the state of the RAM.

- **How it works:** When you hibernate, the contents of the RAM (memory) are written to the **Root EBS Volume**. When you start it again, the RAM is reloaded from the disk.
- **Benefits:**
  - Much faster boot time than a full Stop/Start.
  - Applications do not need to "warm up" or reload caches.
- **Requirements (Important for Exam):**
  - The Root EBS Volume must be **Encrypted**.
  - Must be enabled at **Launch** (cannot enable on an existing running instance).
  - Root volume must have enough space to store the RAM contents.
  - Not supported for bare metal instances or instances with Instance Store.
- **Lifecycle:**
  - **Instance ID:** Retained.
  - **Private IP:** Retained.
  - **Elastic IP:** Retained.
  - **Public IP (Standard):** Lost (same as Stop/Start).
- **Maximum Duration:** You cannot keep an instance in hibernation for more than 60 days.